PROBLEM STATEMENT

SOFTWARE :

1. AI-Powered Crop Disease Detection Develop a system where farmers upload crop images and get disease detection, treatment suggestions, and nearby supplier recommendations.

2. Local Service Marketplace for Small Businesses Design a platform connecting local service providers (plumbers, electricians, etc.) with customers, including booking, reviews, and pricing transparency.

3. Traffic Violation Detection System Develop a system using video/image processing to detect violations (no helmet, signal jump, overspeed) and generate automated reports.

4. Job Matching Platform for Blue-Collar Workers Build a simple, multilingual platform connecting daily wage workers with employers, including location-based matching and skill tagging.

5. Disaster Relief Resource Management System Create a platform to manage distribution of food, medicine, and shelter during disasters with real-time tracking and need prioritization.

6. Open innovation (SDG'S)

HARDWARE :

1. Design a single-feed antenna that operates at both 2.4 GHz and 5 GHz bands.
Requirements:
Two operating frequencies: 2.4 GHz & 5 GHz
Constraints:
S11 ≤ -10 dB at both frequencies
Clear separation between bands
Gain ≥ 4 dB
Submission:
S11 showing dual resonance
Radiation pattern at both frequencies
Design justification

2. Improve the bandwidth of a conventional 2.4 GHz patch antenna using structural modifications.
Requirements:
Base frequency: 2.4 GHz
Constraints:
Bandwidth ≥ 200 MHz
Gain ≥ 4 dB
Submission:
Before vs after S11 plots
Bandwidth calculation
Brief explanation

3. Design a reconfigurable antenna capable of switching between two frequency bands using simulation-based switching.
Requirements:
At least 2 frequencies
Constraints:
S11 ≤ -10 dB in both states
Clear frequency reconfiguration
Submission:
S11 plots (both states)
Switching explanation

4. Design an antenna array capable of steering its beam electronically.
Constraints:
Beam steering ≥ ±30°
Side lobes minimized
Submission:
Radiation patterns at different angles
Phase vs angle explanation

5. Design an antenna covering a wide frequency band for high-speed communication.
Requirements:
Frequency range: 3.1–10.6 GHz
Constraints:
S11 ≤ -10 dB across band
Stable radiation pattern
Submission:
Wideband S11 plot
Gain vs frequency

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